When Fair Baselines Erase the Quantum Edge: A Matched-Capacity Study of Quantum Reservoir Computing at NISQ Scale

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June 13, 2026

Abstract

Quantum reservoir computing (QRC) is widely reported to outperform classical methods on time-series tasks, but many such comparisons rely on classical baselines that are under-tuned, mismatched in capacity, or compared on a metric that flatters the quantum model. We revisit two representative QRC advantage mechanisms under deliberately fair conditions at noisy intermediate-scale quantum (NISQ) sizes ($q \leq 11$, statevector simulation): (i) a two-point correlator readout applied to genuinely quadratic multivariate dynamics, and (ii) feedback-driven QRC on nonstationary regime-switching signals. For each we introduce the control the QRC literature typically omits: a *capacity-matched* classical model (a degree-2 polynomial whose feature set spans the same quadratic interactions the correlator readout exposes) in case (i), and a *tuning-matched* classical recurrent network (an echo-state network optimised with the identical validation budget) in case (ii). Across both, the quantum reservoir's apparent advantage disappears once the comparison is fair: the correlator QRC carries no information beyond the polynomial basis (concatenation improves it by at most 3.9%, consistent with mild regularisation), and feedback QRC—although a genuinely live mechanism that lifts a useless open-loop reservoir to predictive performance—still trails a same-budget echo-state network by 14% in normalised error. We argue that the methodological controls, not the negative outcomes, are the contribution: they provide a reusable template for honest QRC benchmarking and a bounded, evidence- backed map of where the quantum reservoir does not help at currently simulable scales. All results are deterministic and reproduce bit-for-bit.

1 Introduction

Reservoir computing trains only a linear readout on top of a fixed nonlinear dynamical system, which makes it an attractive near-term application of quantum hardware: a fixed quantum reservoir avoids the trainability pathologies of variational circuits while supplying a high-dimensional nonlinear feature map. A growing body of work reports that quantum reservoir computing (QRC) outperforms classical reservoirs and recurrent networks on chaotic, financial, and biomedical time series.

A recurring weakness in these reports is the classical baseline. When a quantum model is compared against an echo-state network (ESN) or a polynomial readout, the strength of the conclusion depends entirely on whether the classical model was given an equal chance: the same feature-space capacity, the same hyperparameter-tuning budget, the same causal evaluation protocol, and a metric that does not implicitly favour the quantum side. In practice these conditions are often not met. Baselines are reported with default settings while the quantum model is tuned; "classical reservoir" refers to a different architecture than the quantum one;

efficiency is measured as simulation wall-clock time rather than any hardware-relevant quantity; and directional accuracy is reported without the persistence control that makes it meaningful.

This paper asks a deliberately narrow question: *when the classical baseline is matched in capacity and tuning, does the quantum reservoir's reported advantage survive?* We study two representative advantage mechanisms at NISQ scale and find that it does not. Our contribution is not the negative outcome per se but the two controls that produce it—a capacity-matched classical model and a tuning-matched classical recurrent network—together with a correctness-gated, deterministic simulation harness. We state the scope plainly: this is a simulation study at sizes that classical hardware can represent exactly ($q \leq 11$); the claim is "no advantage here, under fair comparison," not "no advantage is possible."

2 Methods

2.1 Common harness and fairness controls

All experiments share a single evaluation harness with the following properties, several of which are the fairness controls whose absence we are critiquing.

Causal, leakage-free evaluation. Each series is split chronologically into training, validation, and test segments. Every preprocessing step (standardisation) and every hyperparameter is fit on the training/validation portion only and applied causally; the test segment is touched once, for the final reported number.

Validation-only tuning, applied equally. All free hyperparameters—ridge penalty for every method, quantum evolution time τ, feedback gain k_{fb}, and the ESN spectral radius and leak rate—are selected by validation error. Crucially, the classical baselines receive the *same* tuning budget as the quantum model. This is the single most important control and the one most often missing.

Matched feature dimension. Where a quantum and a classical reservoir are compared, their readout feature dimensions are matched so that the comparison is not confounded by readout width.

Correctness gates. Before any result is generated, an automated self-test verifies: the reservoir state is physical ($\mathrm{Tr}\,\rho = 1$ and all measured expectations lie in $[-1, 1]$); the two-point readout computed from the density-matrix diagonal agrees with the explicit operator expectation to machine precision; finite-shot estimates converge to the exact values as the shot count grows; and, for the feedback model, that the feedback path is wired correctly (with gain zero it reproduces the open-loop reservoir to machine precision) and is strictly causal (perturbing a future input leaves earlier features unchanged). Every reported number is read back from disk and checked against a content hash; all results in this paper are deterministic and reproduce bit-for-bit.

Metrics. We report normalised root-mean-square error (NRMSE, normalised by the per-target standard deviation, so that NRMSE = 1 is the trivial mean predictor) and, for directional tasks, mean directional accuracy (MDA) reported alongside a persistence baseline.

2.2 Quantum reservoirs

The quantum reservoir is a fully connected transverse-field Ising system,

$$H = \sum_{i<j} J_{ij}\, X_i X_j + v \sum_i Z_i, \qquad J_{ij} \sim \mathcal{U}[0,1],\ v = 1, \tag{1}$$

fixed after random initialisation. Classical inputs are angle-encoded via single-qubit R_Y rotations; the state evolves for time τ; and the readout consists of single-qubit expectations $\langle Z_i \rangle$ and, where indicated, two-point correlators $\langle Z_i Z_j \rangle$, which are estimable from the same measurement record. A recurrent variant carries memory across timesteps by partial-tracing the input qubits between encodings. Because Z_i and $Z_i Z_j$ are diagonal in the computational basis, all correlators are obtained from the state-diagonal in a single contraction, verified equal to the explicit operator expectation (Section 2).

3 Case study I: a correlator readout adds nothing beyond a capacity-matched polynomial

3.1 Setup

A frequently proposed route to quantum expressivity is to enrich the readout with two-point correlators $\langle Z_i Z_j \rangle$, which capture pairwise quantum correlations and grow the feature space quadratically. The natural question a fair comparison must ask is whether these features carry information that an explicit classical quadratic model does not already have.

We test this on a controlled testbed of $N = 5$ coupled Hénon maps—bounded, genuinely quadratic, multivariate dynamics with a tunable cross-asset coupling c. We compare three models on one-step prediction (all with the same windowed ridge readout and causal protocol): the tuned correlator QRC ($q = 8$, readout tuned over τ, encoding scale, and single- vs. two-reservoir ensemble); a degree-2 polynomial model (Poly2) whose features are all pairwise products of the windowed inputs—the *capacity-matched* classical control, deliberately given *more* features than the QRC; and the concatenation Poly2 $\oplus$ QRC, which tests directly whether the quantum features add anything to the polynomial basis. Observation noise of standard deviation 0.1 is added so that no model can solve the map exactly through its generating polynomial. The concatenation ablation—running the classical and quantum feature sets together and asking whether the union beats the classical part alone—is exactly the control omitted by typical hybrid-architecture reports.

3.2 Results

Table 1 reports the verified numbers. At the one-step horizon the Poly2 control beats the tuned QRC outright at every coupling (e.g. NRMSE 0.327 vs. 0.713 at $c = 0.1$), and at the strongest coupling the tuned QRC degrades past the trivial-mean predictor (NRMSE = 1.45) while the classical models remain predictive. Decisively, concatenating the quantum features onto Poly2 improves on Poly2 alone by between 0.1% and 3.9%—the magnitude expected from a few extra mildly-regularising features, not from complementary signal. The same verdict holds on mean directional accuracy: Poly2 reaches MDA ≈ 0.96 (persistence ≤ 0.18), the quantum model is no better, and the concatenation does not exceed Poly2. The coupling axis, which one might expect to favour the joint quantum encoding, instead makes the QRC relatively worse.

The interpretation is unambiguous: on dynamics that are genuinely quadratic—the regime most favourable to a two-point correlator readout—the quantum reservoir contributes no information that an explicit degree-2 polynomial does not already hold. The correlator readout helps the QRC relative to a single-point readout, but that gain does not transfer into an advantage over the classical model whose capacity it was meant to match.

Table 1: Case study I (one-step, observation noise 0.1): correlator QRC vs. the capacity-matched POLY2 control and their concatenation, on coupled Hénon dynamics at three couplings c. Lower NRMSE is better; NRMSE < 1 is predictive. "cat" is POLY2 $\oplus$ QRC. All values verified from disk.

c	QRC	POLY2	cat	cat vs. POLY2
0.0	0.712	0.353	0.344	+2.4%
0.1	0.713	0.327	0.323	+1.3%
0.2	1.450	0.529	0.509	+3.9%

4 Case study II: feedback is a live mechanism but loses to a tuning-matched ESN

4.1 Setup

A distinct advantage claim concerns *feedback-driven* QRC, in which the measurement outcome at one step modulates the reservoir drive at the next, making the effective dynamics path-dependent rather than a fixed function of a finite input window. Path-dependence is something a fixed-window polynomial cannot reproduce—but it is precisely what a classical recurrent network also provides. The honest baseline for a feedback reservoir is therefore not a polynomial but a recurrent network, the echo-state network (ESN), and the test is whether *quantum* recurrence beats *classical* recurrence at equal tuning.

We use a nonstationary task built so that recurrence is necessary: a hidden two-state Markov process switches the sign of a first-order autoregressive coefficient, so each regime is individually predictable but the regime must be inferred online from the observable history. A correctness gate confirms the task genuinely requires regime inference (an oracle given the hidden state beats a fixed-window linear model, with the gap growing as the switching rate rises). We implement feedback as an additive, saturating term on the input drive angle, $\phi_t = x_t + k_{\text{fb}} \tanh(\bar{m}_{t-1})$, where $\bar{m}_{t-1}$ is the mean measurement at the previous step; $k_{\text{fb}} = 0$ recovers the open-loop reservoir exactly. We compare, over three data seeds $\times$ three reservoir seeds, the tuned feedback QRC, its open-loop counterpart, a tuning-matched ESN (spectral radius and leak rate selected on the same validation budget), POLY2, and a linear model.

4.2 Results

Feedback is a genuine, live mechanism. The open-loop reservoir is useless on this task (NRMSE $\approx$ 1.06, worse than the mean predictor); closing the loop lifts it to predictive performance, an improvement of 10–19% across switching rates (Table 2, upper). This is, to our knowledge, the clearest case in our study of a quantum reservoir mechanism doing something its open-loop version cannot.

The mechanism being live, however, does not make it advantageous. Under the fair 3×3-seed comparison (Table 2, lower), the tuning-matched ESN attains NRMSE $= 0.662 \pm 0.029$ against the feedback QRC's 0.756 ± 0.072—an ESN advantage of 14%. The feedback QRC in fact trails every classical baseline, including a plain windowed linear model (0.692). On directional accuracy all models cluster near 0.76, far above the 0.24 persistence floor but indistinguishable across methods. The conclusion is that quantum feedback here reconstructs the recurrent memory that a classical recurrent network already provides, and does so slightly less efficiently; closing the loop rescues the quantum reservoir up to, but not past, the classical pack.

Table 2: Case study II. Upper: feedback vs. open-loop QRC across switching rates p_{sw} (one seed), showing feedback is a live mechanism. Lower: the fair test at $p_{\mathrm{sw}} = 0.05$ over 3×3 seeds, mean $\pm$ s.d., one-step NRMSE and MDA (persistence MDA = 0.24). The tuning-matched ESN wins.

Upper: feedback vs. open-loop			
p_{sw}	open-loop	feedback	gain
0.02	1.007	0.874	+13.2%
0.05	1.018	0.824	+19.1%
0.10	1.032	0.925	+10.5%

Lower: fair comparison, 3×3 *seeds*		
method	NRMSE	MDA
ESN (tuned)	0.662 ± 0.029	0.764 ± 0.037
Linear	0.692 ± 0.018	0.775 ± 0.033
Poly2	0.729 ± 0.036	0.773 ± 0.038
Feedback QRC	0.756 ± 0.072	0.760 ± 0.035
Open-loop QRC	1.061 ± 0.025	0.759 ± 0.042

5 Discussion

Two mechanisms, two representative advantage claims, one outcome: under a capacity-matched classical control in the first case and a tuning-matched classical recurrent control in the second, the quantum reservoir's reported edge does not survive. We emphasise three points.

First, the controls are the contribution. The concatenation ablation (Poly2 $\oplus$ QRC vs. Poly2) directly measures whether quantum features add information to a classical model of matched capacity, and the same-budget ESN measures whether quantum recurrence beats classical recurrence on equal footing. These are inexpensive to run and decisive in interpretation, yet they are routinely omitted; including them changes the conclusion.

Second, "live" is not "advantageous." The feedback reservoir genuinely transforms a useless open-loop model into a predictive one. A study that compared feedback QRC only against open-loop QRC—or against an untuned ESN—would have reported a success. The advantage evaporates only against a classical recurrent model tuned as hard as the quantum one.

Third, the scope is bounded and we state it as such. These are statevector simulations at $q \leq 11$, sizes a classical computer represents exactly, on tasks with low-dimensional classical inputs. We make no claim about larger systems, genuinely quantum-state inputs, or non-Clifford resources, where the relevant separations may behave differently. Our claim is the narrow, falsifiable one our experiments support: at currently simulable scales, with fair baselines, these two QRC advantage mechanisms do not beat their classical counterparts.

6 Conclusion

We have shown, under deliberately fair comparison, that two representative quantum-reservoir advantage mechanisms—a two-point correlator readout and a feedback-driven reservoir—do not outperform capacity- and tuning-matched classical baselines at NISQ scale. The methodological template we use, a capacity-matched classical control plus a same-budget recurrent control inside a correctness-gated, deterministic harness, is a reusable standard for QRC benchmarking. We offer the resulting map not as a verdict on quantum reservoir computing but as a marker of where, today, the quantum part earns its keep and where it does not.

Reproducibility. All experiments are deterministic; each reported value was regenerated from a fresh run and verified bit-for-bit against the archived result via content hash. Code and logged results are available from the author.